$$C_V = \frac{dU}{dT}$$

$dU = C_V\, dT$

For a finite change, we have

$\Delta U = C_V\, \Delta T$

$$\Delta U = C_V\, (T_2 - T_1)$$

and for n moles of an ideal gas we get

$\Delta U = n\, C_v (T_2 - T_1)$ ---------- (7.25)

Similarly for *n* moles of an ideal gas we get

$\Delta H = n\, C_P\, (T_2 - T_1)$------ (7.26)

Problem 7.3

Calculate the value of ΔU and ΔH on heating 128.0 g of oxygen from 0° C to 100^0 C. C_V and C_P on an average are 21 and 29 J mol^{-1} K^{-1}. (The difference is 8Jmol^{-1} K^{-1} which is approximately equal to R)

Solution.

We know

$\Delta U = n\, C_v\, (T_2 - T_1)$

$\Delta H = n\, C_P\, (T_2 - T_1)$

Here $n = \frac{128}{32} = 4$ moles ;

$T_2 = 100^0$ C =373K; $T_1 = 0^0$ C = 273K

$\Delta U = n\, C_v\, (T_2 - T_1)$

$\Delta U = 4 \times 21 \times (373 - 273)$

$\Delta U = 8400$ J

$\Delta U = 8.4$ kJ

$\Delta H = n\, Cp\, (T_2 - T_1)$

$\Delta H = 4 \times 29 \times (373 - 273)$

$\Delta H = 11600$ J

$\Delta H = 11.6$ kJ

Evaluate Yourself -2

Calculate the amount of heat necessary to raise 180 g of water from 25^0 C to 100^0 C. Molar heat capacity of water is 75.3 J mol^{-1} K^{-1}

7.7 Measurement of ΔU and ΔH using Calorimetry.

Calorimeter is used for measuring the amount of heat change in a chemical or physical change. In calorimetry, the temperature change in the process is measured which is inversely proportional to the heat change. By using the expression $C = q/m\Delta T$, we can calculate the amount of heat change in the process. Calorimetric measurements are made under two different conditions

i) At constant volume (q_V)

ii) At constant pressure (q_p)

(A) ΔU Measurements

For chemical reactions, heat evolved at constant volume, is measured in a *bomb calorimeter*.

11th Std Chemistry 186-228.indd 202 06-01-2021 15:46:13